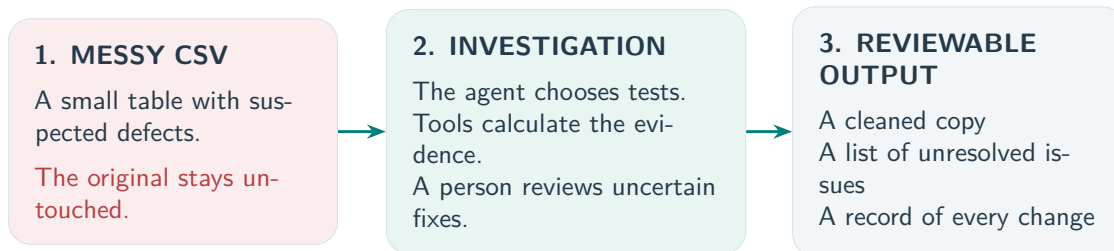


01 / THE BIG PICTURE

What are we building?

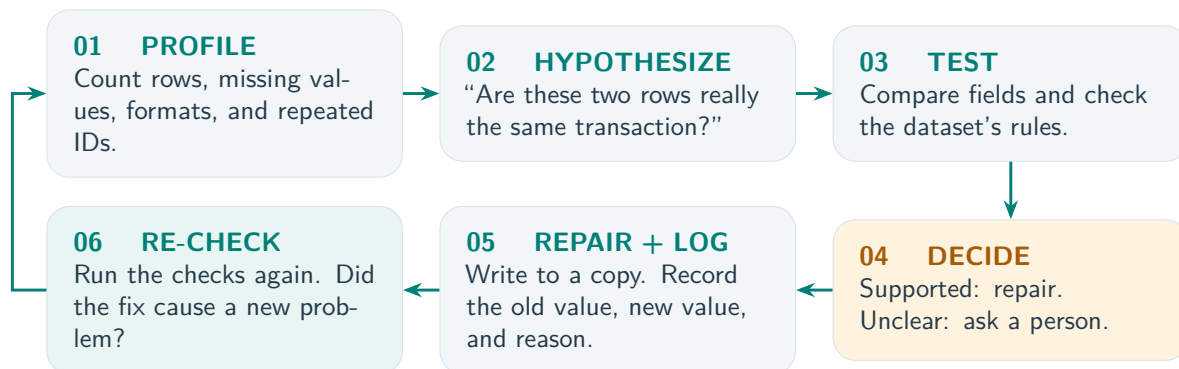
An AI data detective that must justify every repair.

Imagine a spreadsheet with repeated sales, missing quantities, and strange values. Cleanroom Detective investigates the problems, repairs what the evidence supports, and asks a person about anything ambiguous.



What makes it an agent?

A fixed script follows the same recipe every time. Here, the agent chooses its next test from what it finds. If the evidence is weak, it investigates further or asks a human.



The core promise

Every repair has a reason you can inspect. A surprising value is a clue to investigate; it is not, by itself, proof that the data is wrong.

Status: this document explains the proposed prototype. The product has not yet been built or evaluated.

02 / A WORKED EXAMPLE

Watch one investigation

Five cafe sales. Three decisions. Different kinds of evidence.

Illustrative data only. The examples below are expected behavior, not results from a working system. All prices are in USD.

Row	Sale ID	Item	Qty.	Unit price	Total
1	T101	Coffee	2	3.00	6.00
2	T101	Coffee	2	3.00	6.00
3	T102	Tea	?	2.50	7.50
4	T103	Sandwich	1	8.00	8.00
5	T104	Coffee	20	3.00	60.00

The rules must be known first

In this example, each Sale ID identifies exactly one sale row. Quantity is a positive whole number, prices and totals are trusted, and there are no taxes, discounts, or refunds. These are supplied data rules, not assumptions the AI may invent.

A / Two identical rows: a supported repair

Rows 1 and 2 have the same ID *and* matching fields. Under the unique-ID rule, row 2 is an extra copy. Keep row 1; remove row 2 from the cleaned copy.

Evidence: identical record + a rule saying that ID must be unique. If an ID can span several line items, this rule no longer justifies deletion.

B / Missing quantity: recover it from trusted fields

For row 3, the known accounting rule gives:

$$\text{quantity} = \frac{\text{total}}{\text{unit price}} = \frac{7.50}{2.50} = 3.$$

Set the quantity to **3**; then verify that $3 \times 2.50 = 7.50$. Without the stated rules, this would be a proposal for review, not an automatic fix.

C / Twenty coffees: unusual does not mean incorrect

Row 5 is large, but $20 \times 3.00 = 60.00$ is consistent. It could be a group order. **Keep 20 unchanged.** A person can check the receipt if clarification is needed. The agent must not replace it with a typical value simply because it is unusual.

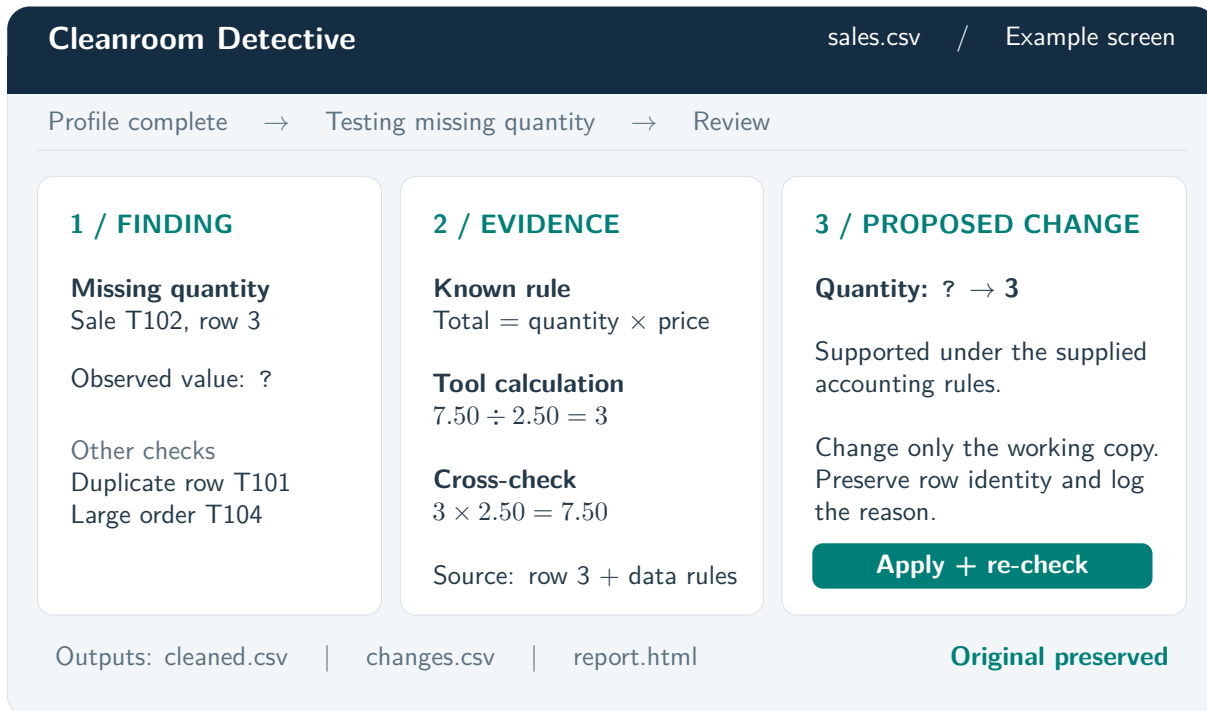
Cleaned example: 4 rows ■ **T102 quantity = 3** ■ **T104 quantity stays 20**

The change log records row 2's removal and row 3's replacement, with reasons. The original five-row file is preserved so each change can be undone.

03 / WHAT YOU WOULD SEE

The investigation screen

One place to see the issue, the evidence, and the proposed action.



Design mockup, not an application screenshot. An ambiguous case would show a human-review action instead of an automatic repair. No invented accuracy or confidence percentages.

Who does what?

The agent and its tools

Agent: chooses what to investigate, selects a tool, interprets the result, and chooses repair, more testing, or review.

Tools: count, compare, calculate, apply a patch to a copy, and run checks again.

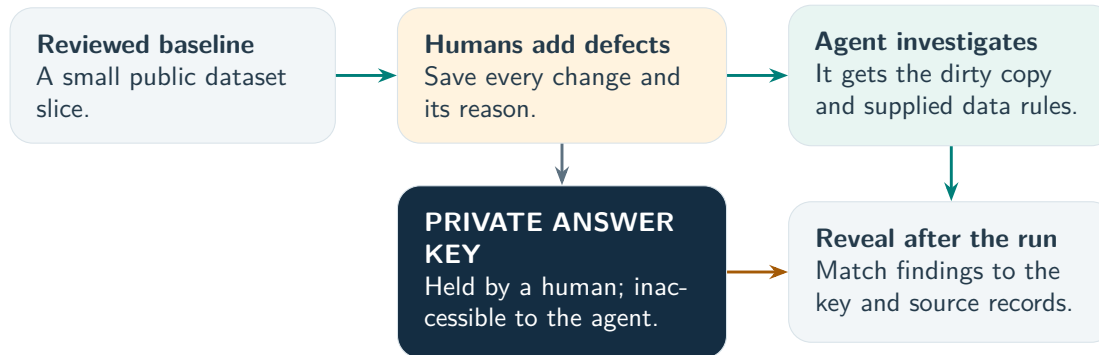
The human teammate

Supplies trustworthy data rules, resolves ambiguous cases, and checks the result against the original source. Can reject a proposed repair and explain why. A second AI agreeing is not a human check.

04 / HOW WE PROVE IT WORKS

The reveal is the demo

Humans know the planted errors. The agent does not.



CAUGHT

A real defect was identified. Check repair correctness separately.

MISSED

A real defect was not identified. Track unresolved repairs separately.

FALSELY FLAGGED

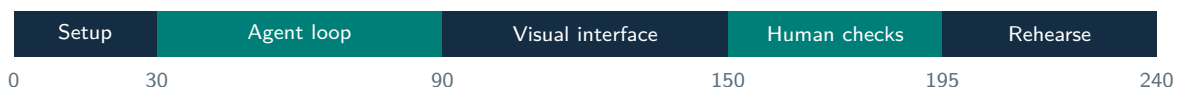
A valid value was incorrectly called an error; a human checks it.

Humans hold both the baseline and key until the agent's findings are frozen. Score only verified cases; the key does not prove that all other data is flawless. Show an actual agent mistake, never a fabricated one.

A two-minute presentation

- 0:00–0:20** Show the messy file and explain the hidden answer key.
- 0:20–0:55** Give one audit goal. Show the agent choosing and running tests.
- 0:55–1:20** Open one finding; inspect the evidence and the actual repair.
- 1:20–1:45** Reveal caught, missed, and falsely flagged cases. Explain a real human correction.
- 1:45–2:00** Export the cleaned copy and change ledger. Show what remains unresolved.

Keep the four-hour build small



Estimated minutes, not a measured schedule. Setup includes verifying the model/tool connection and preparing a reviewed baseline.

MVP boundary: one CSV, five defect types, a small tool set, one review screen, and three export files. Start with duplicates and missing values; add invalid dates, inconsistent labels, and documented unit mismatches. Keep a valid outlier as a control.

Source: 2026 - Hackathon Challenge Problems.pdf, challenge 9, page 7. The challenge requires profile/check/fix/re-profile, reasons for fixes, and human verification. The interface, example data, and schedule in this explainer are proposed designs.